# Grace Wou

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## EDUCATION

**University of Michigan, Bachelor of Science in Engineering** **Aug. 2024 – May 2028**
*Mechanical Engineering Major, Robotics Concentration* *Ann Arbor, MI*

## EXPERIENCE

**MedLaunch** **Sep. 2024 – Present**
*Internal Communications Chair* *Ann Arbor, MI*

- Engaging in year-long biodesign projects developing assistive technologies for community partners with disabilities.
- Developed a custom sports vest for an amputee soccer goalkeeper by creating sewing patterns that optimize mobility and structure, researching fabric materials, and rapid prototyping the vest through sewing and functional testing.
- Designed and built an adaptive wheelchair device by implementing quick-attach wheelchair mechanisms, evaluating surface concavities for optimal hitting surface and power, and testing solutions with athletes in game-like scenarios.

**Boston Scientific** **May 2026 – Aug. 2026**
*Process Development Intern* *Arden Hills, MN*

- Designed and developed a custom PCB fixture from concept through CAD, rapid prototyping, and 3D printing, increasing usable board capacity by 57% while maintaining support for capacitor terminal pins and improving operator usability.
- Led a process improvement DMAIC investigation targeting capacitor electrolyte weight-gain variability historically responsible for 7% rework, utilizing Ishikawa diagrams, statistical analysis, microscopy inspection, and controlled experimental testing to identify critical process drivers.

**Harbinger Motors** **May 2025 – Aug. 2025**
*Quality Engineering Intern* *Orange County, CA*

- Reduced chassis defects by 48% by applying quality engineering methodologies, including 4 Diamonds and Kaizen.
- Conducted root cause analysis of a wheel bearing NCR by utilizing CAD models and work instructions, transitioning from force-based to distance-based pressing to improve consistency and capability of the wheel bearing press.
- Designed and fabricated precision gauges for wheel bearing measurements with CAD and 3D printing, reducing frequency of inspection processes by 90%.
- Operated 3D printing equipment to fabricate 10+ prototype components for design iterations.

## PROJECTS

**Plants vs. Zombies** **Sep. 2026 – Present**
*Design and Manufacturing II*

- Designed and machined a 4-bar linkage in SolidWorks to position a flashlight across four photosensor locations, and validated inertia, torque, and transmission ratio with MSC ADAMS simulation to select the optimal motor-to-linkage configuration.
- Integrating an Arduino-controlled closed-loop DC motor system with limit switches and toggles to move the mechanism between four target positions fast enough to return targets before they reach the front limit switch.

**Guitar Pedal** **Sep. 2026 – Present**
*Personal Project*

- Building a functional guitar pedal circuit, applying principles of circuit design, soldering, and electronic troubleshooting.
- Designing and fabricating a custom pedal enclosure with CAD and 3D printing, incorporating component placement, mounting features, and accessibility for assembly.

**Robotic Machine Player** **Sep. 2025 – Dec. 2025**
*Design and Manufacturing I*

- Engineered a rack-and-pinion linear motion system and custom forklift mechanism in SolidWorks, producing detailed CAD models, assemblies, and fabrication drawings.
- Manufactured 10+ parts using a manual mill and lathe while achieving tolerances of 0.001".
- Performed mechanical assembly, electrical wiring, and soldering to integrate hardware subsystems into functioning robot.

## TECHNICAL SKILLS

**Languages**: Python, Java, C++, MATLAB
**CAD/Fabrication**: SolidWorks, CATIA, 3D Printing, Mill, Lathe, Waterjet, Soldering
**Tools**: NetSuite, Jira, Microsoft Excel, Minitab, MSC Adams